Instrumental Methods and Applications

5.1 INTRODUCTION

Instrumental methods (or) Analytical technique (or) Spectroscopy is one of the most powerful tool available for the study of atomic and molecular structure, and is used in the analysis of a most of the samples.

Spectroscopy deals with the study of interaction of electromagnetic radiation with the matter. During the interaction, the energy is absorbed (or) emitted by the matter. The measurement of this radiation frequency (absorbed (or) emitted) are made using spectroscopy.

5.2 TYPES OF SPECTROSCOPY

The study of spectroscopy can be carried out under the following headings

1. Atomic spectroscopy.
2. Molecular spectroscopy.

1. Atomic spectroscopy

It deals with the interaction of the electromagnetic radiation with atoms. During which the atoms absorb radiation and gets excited from the ground state electronic energy level to another.

2. Molecular spectroscopy

It deals with the interaction of electromagnetic radiation with molecules. This results in transition between rotational, vibrational and electronic energy levels.

Table 5.1 Differences between molecular spectra and atomic spectra

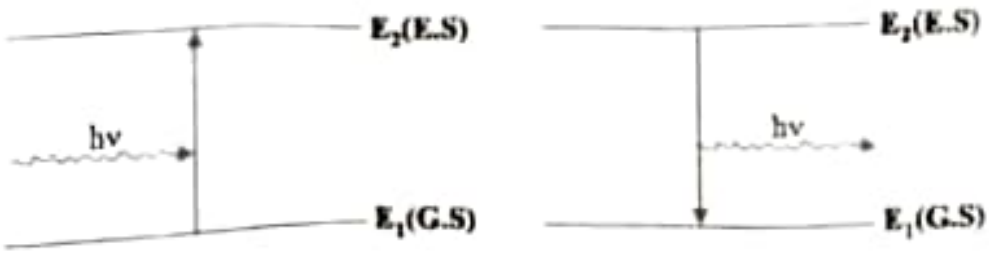
Sl. No.	Atomic spectra	Molecular spectra
1	It occurs from the interaction of atoms + electromagnetic radiation.	It occurs from the interaction of molecules + electromagnetic radiation.
2	Atomic spectra is a line spectra.	Molecular spectra is a complicated spectra.
3	It is due to electronic transition in an element.	It is due to virbrational, rotational and electronic transition in a molecule.

5.3 SPECTRUM

How does a Spectrum arise?

1. Absorption spectrum

Consider a molecule having only two energy levels E_1 and E_2 as shown in the figure. 5.1



(a) Absorption spectrum **(b) Emission spectrum**

Fig 5.1 Spectrum

When a beam of electromagnetic radiation is allowed to fall on a molecule in the ground state, the molecule absorbs photon of energy $h\nu$ and undergoes a transition from the lower energy level to the higher energy level. The measurement of this decrease in the intensity of radiation is the basis of absorption spectroscopy. The spectrum thus obtained is called the **absorption spectrum**. (Fig 5.1. a)

2. Emission spectrum

If the molecule comes down from the excited state to the ground state with the emission of photons of energy $h\nu$, the spectrum obtained is called ***emission spectrum***. (Fig 5.1.b).

5.4 ABSORPTION OF RADIATION

When electromagnetic radiation is passed through a matter, the following changes occur.

1. As the photons of electromagnetic radiations are absorbed by the matter, electronic transition, vibrational changes (or) rotational changes may occur. After absorption, molecules get excited from the ground state to excited state. Then they liberate energy quickly in the form of heat (or) re-emit electromagnetic radiation.
2. But in some cases, the portion of electromagnetic radiation, which passes into the matter, instead of being absorbed may be scattered (or) reflected (or) re-emitted.
3. When the electromagnetic radiation is absorbed (or) scattered, it may undergo changes in polarisation (or) orientation.
4. In some cases the molecules absorbs radiation and get excited.

(a) Fluorescence

If the excited molecules re-emits the radiation almost instantaneously (within 10^{-8} seconds), it is called fluorescence.

(b) Phosphorescence

If the excited molecules re-emits the radiation after sometime (slowly), it is called phosphorescence.

5.4.1 Factors affecting absorbance

The fractions of photons being absorbed by the matter depends on,

1. ***The nature of the absorbing molecules.***
2. ***The concentration of the molecules.***

   If the concentration of the molecules are more, the absorbed photons will be more.

3. ***The length of the path of the radiation through the matter.***

   If the length of the path is long, the larger number of molecules are exposed and hence greater the photons will be absorbed.

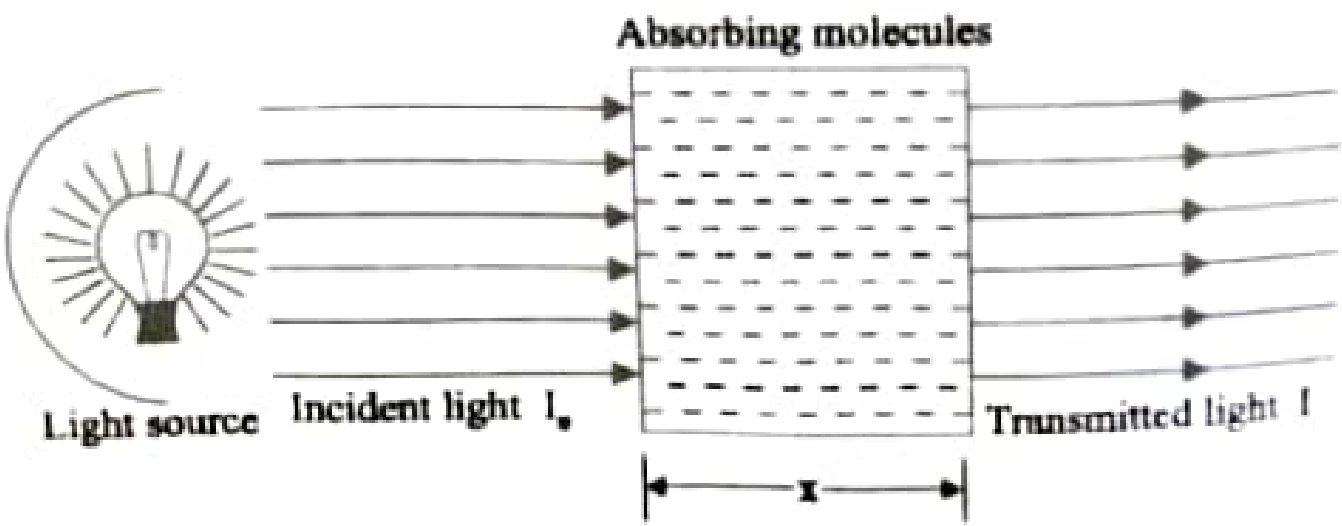
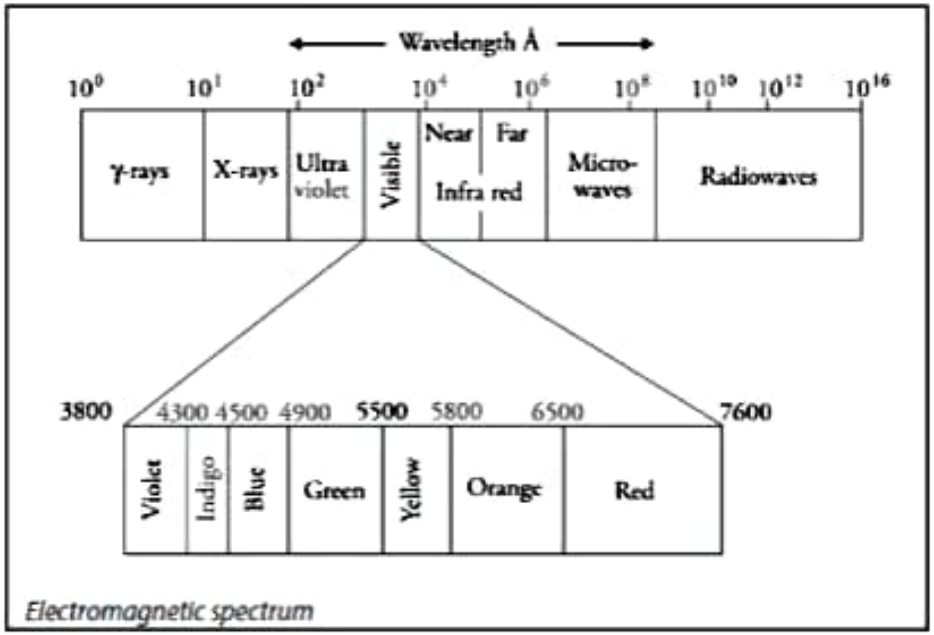


Fig. 5.2 Absorbance of photons by the matter

EMR radiation :

Electromagnetic Spectrum: An electromagnetic spectrum is obtained when all types of electromagnetic radiations are arranged in the order of increasing wavelength or decreasing frequencies (Fig.).



Electromagnetic spectrum

1. Visible light lies in the wavelength range 3800–7600 Å.The region of 3800 Å corresponds to violet colour and that of 7600 Å corresponds to red colour.
2. If the wavelength is less than 3800 Å, the radiation is called ultraviolet radiation
3. if it is greater than 7600 Å, it is called infrared radiations.
4. Both infrared and ultraviolet radiations are not visible to the human eye.
5. Beyond infrared region, on the side of increasing wavelength are microwaves. Beyond microwaves, lie the radio waves which have the highest wavelength.

The arrangement of different types of electromagnetic radiation in increasing order of their wavelength is Cosmic rays < g rays < X rays < Ultraviolet rays < Visible rays < Infrared rays < Microwaves < Radiowaves

BEER-, LAMBERTs Law:-

BEER Law:-

Absorbance is directly propotional to Concentration $\boxed{A \propto C}$

LAMBERT's Law:

Absorbance is directly propotional to Path length

$$\boxed{A \propto l}$$

BEER - LAMBERTs Law:-

According to this law when a beam of mono Chromatic Solution is passed through" a solution of an absorbingce, The rate of decrease intensity of radiation "dl" and Thikness of absorbing solution 'dx' is propotional To intensity of Light 'I" as well as concentration of solution

Mathematically Represented

$$-\frac{dI}{dl} \propto C I$$

$$-\frac{dI}{dl} = KcI$$

$$-\frac{dI}{I} = K \cdot c \cdot dl$$

Applying Integration

$$-\int_{I_0}^{I_t} \frac{dI}{I} = Kc \int_0^l dl$$

$$-\left[\ln I\right]_{I_0}^{I} = K \cdot \left[l\right]_0^l$$

Applying limits

$$-\left[\ln I_t - \ln I_0\right] = K \cdot c \cdot l$$

$$-\left(\ln \frac{I}{I_0}\right) = K \cdot c \cdot l$$

$$2.303 \log \frac{I_0}{I_t} = K \cdot c \cdot l$$

$$\log \frac{I_0}{I_t} = \frac{K}{2.303} \cdot c \cdot l$$

$$\boxed{A = \varepsilon \cdot c \cdot l}$$

$$\left[\log \frac{I_0}{I_t} = A\right.$$

$$\left[\frac{K}{2.303} = \varepsilon\right.$$

A = Absorbance

c = Concentration

l = Path length

ε = Molar absorptivity coefficient

Thus The absorbance (A) directly Proportional to Molar Concentration (c) and Path length (l)

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Date

UV Visible Spectroscopy:

5.6.2 *Limitations of Beer-Lambert's law*

1. Beer-Lambert's law is not obeyed if the radiation used is not monochromatic.
2. It is applicable only for dilute solutions.
3. The temperature of the system should not be allowed to vary to a large extent.
4. It is not applied to suspensions.
5. Deviation may occur, if the solution contains impurities.
6. Deviation also occurs if the solution undergoes polymerization (or) dissociation.

Date

UV Visible Spectroscopy:

Def:- U.V Spectroscopy is the Measurment of accumation of beam of light passing through a Sample (or) Other reflection from Sample Surface

Principle Spectroscopy

* when the molecule absorbs UV (or) Visible light Electron get promoted from ground state to the higher state

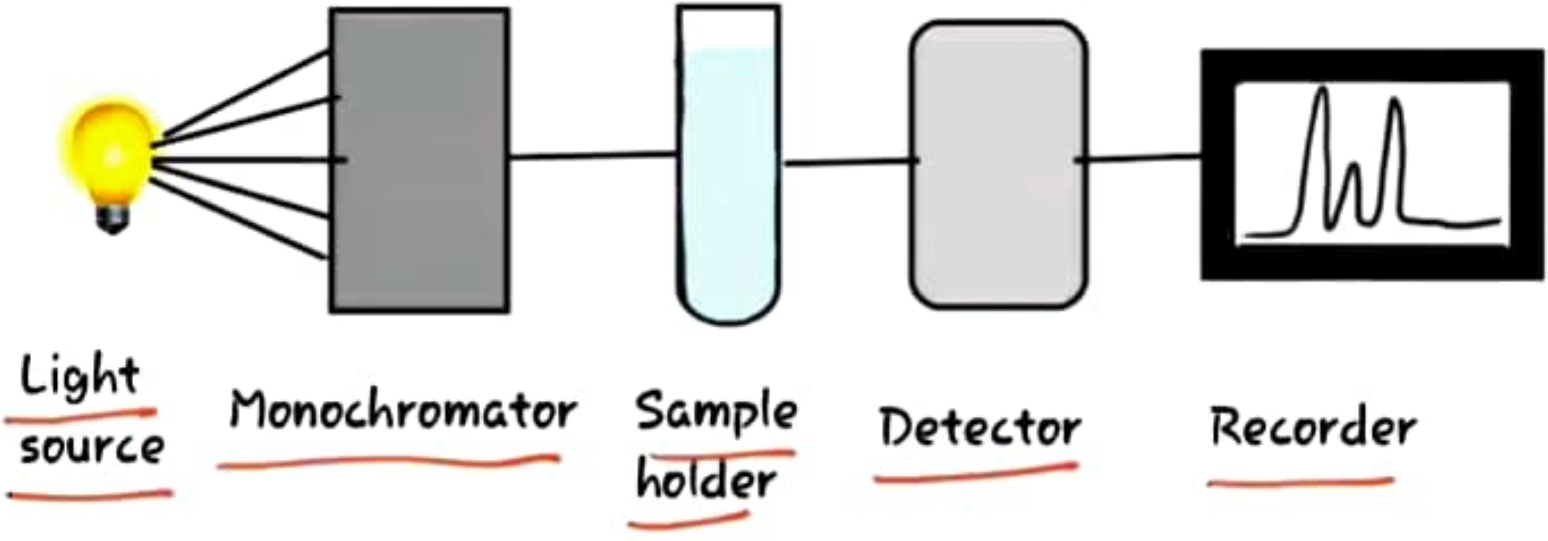
* In the ground state, the Spins of Electrons in Each molecular Orbital Essentially paired

* In the higher Energy state the ~~Electron~~ spin of Electrons are paired is called Singlet state.

* In other hand Spin of Electron excited state are parallal is called Excited Triplet State.

* Excited Triplet state is more stable than Excited Singlet State

* Excited single state Converts to Triplet State Transmission of Energy of light



Instrumentation

Instrumentation

Light Source

1. Duturium Lamp - UV region (200 - 400 nm)
2. Tungstun. halogen lamp - visible region (400 - 800 nm)

2. Monochromator:-

Monochromator is a device used to Resolve wide band of polychromatic light radiation into narrow level band of Monochromatic. radiation

Eg: Filters, Grattings, prisims

3. Sample holder:-

Cuvette is used as Sample holder.

Made up on Quartz

4. Detectors:-

It will Converts light Energy into Electrical Signal That are displayed on read out device.

1. Barrier layer cell
2. photo tube
3. photo Multiplier tube
4. Thermo couple.
5. ~~Bolo~~ Bolt meter

Amplifier & Recorder:-

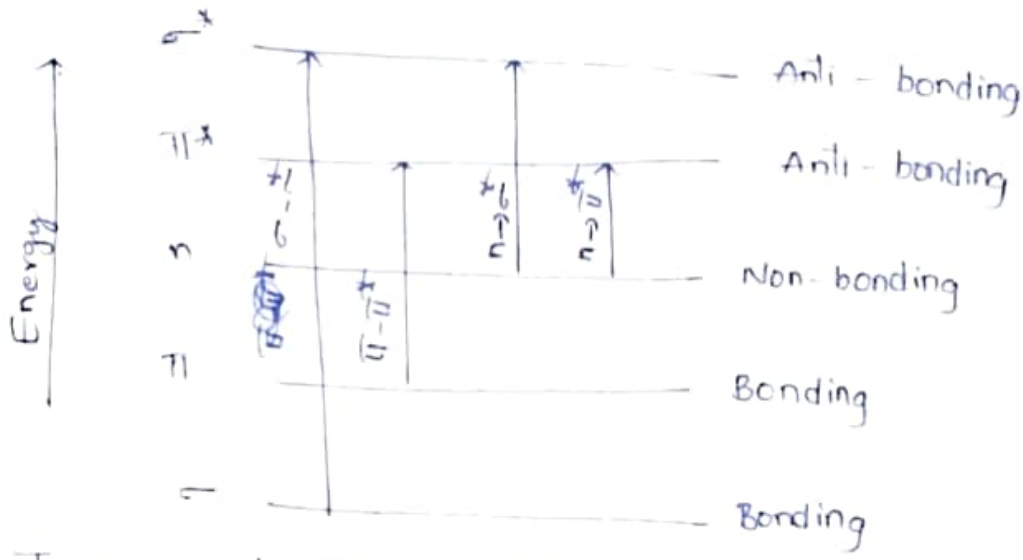
Amplifier Amplifies Signal Coming from detector Chamber and recorder records the Signal which is displayed on read out device

Applications of UV:-

1. Determination Molecular weight of molecule
2. Determination of impurities present in the Sample
3. unknown Concentration of a solution can be determined by spectroscopy
4. Characterisation of aromatic Compound and detection of Conjugation
5. It is used as detector in HPLC

Electronic Transitions:-

When a molecule get excited by the absorption of Electromagnetic radiation in UV-Visible region then its electrons are Transmitted from ground state Excited state or from bonding orbital to anti bonding Orbital. These Transitions are $\sigma - \sigma^*$, $n - \sigma^*$, $n \rightarrow \pi^*$, $\pi - \pi^*$



Types of Electronic Transitions:

1. $\sigma - \sigma^*$ Transition:-

(a) The Transition of an Electron from ~~being~~ bonding sigma orbital (σ) to anti - bonding sigma orbal σ^*, is represented by $\sigma - \sigma^*$ Transition

(b) This type of Transition requires high Energy

(c) $\sigma - \sigma^*$ Transition occures in the Compound, in which all the Electrons are involved in Single bonds and there are no lone pair of Electron

(d) This is observed with Saturated compounds

2. $n \rightarrow \sigma^*$ Transition

(a) Transition from non-bonding orbital (n) to anti bonding Sigma orbital (σ^*), is represented by $n \rightarrow \sigma^*$ Transition

(b) This Type of transition requires less Energy when Compared to $\sigma - \sigma^*$ Transition

(c) This Transition occurs in Saturated Compounds with hetero atom like S, O, N (or) halogens

(d) This absorption band occurs at longer wavelength in near UV region

3. $\pi - \pi^*$ Transition.

(a) Transition from bonding pi orbital (π) to anti bonding pi orbital (π^*). is represented by $\pi - \pi^*$ Transition

(b) the Energy requirment of This Transition is between $\sigma - \sigma^*$ to $n \rightarrow \pi^*$

(c) This type of Transition occurs in any molecule having π- Electron System

(d) This Type of Transition generally shows in unsaturated molecule like alkenes, alkynes, aromatic & Carbonyl Compounds

4. $n \rightarrow \pi^*$ Transition

1. Transition from non bonding molecule (n) orbital to anti bonding ~~sigma~~ pi orbital (π^*), it is represented $n \rightarrow \pi^*$ Transition

(b) This type of Transition requires lowest Energy

(c) The peaks due to this Transition are also Called R- bands

(d) This type of peak Can be Seen in the Compound when non bonding Electron (present in S, O, N or halogens) is present in a compound containing double bond or Triple bond

Ex:- aldehyde, Ketone, etc

Order of Electronic Transitions

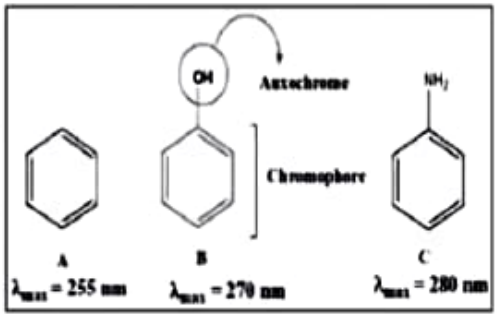
****

$$\boxed{\sigma - \sigma^* >> n \rightarrow \sigma^* > \pi - \pi^* > n \rightarrow \pi^*}$$

Chromophores & Auxochromes :

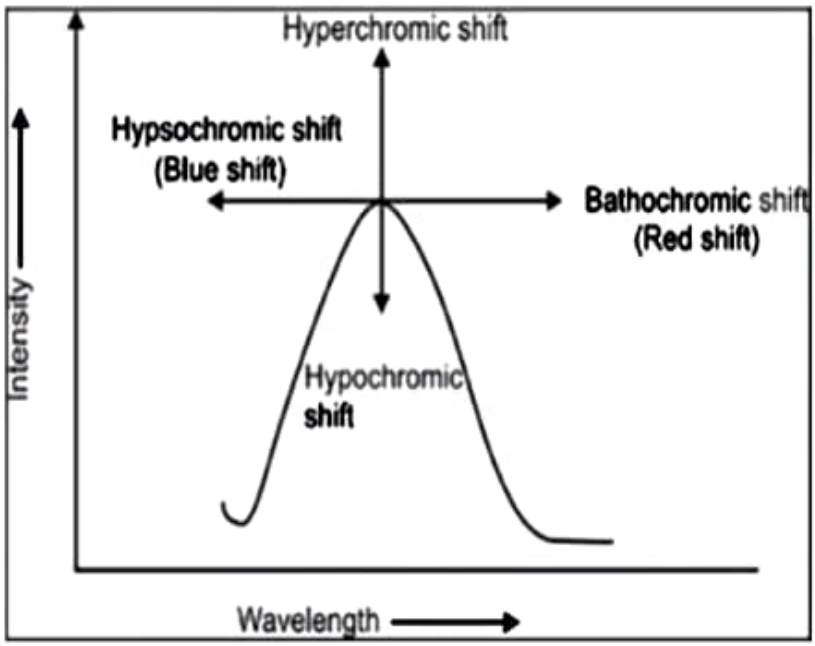
Chromophores: Chromophore is a covalently unsaturated group absorbed electromagnetic radiation in UV-visible region and impart color to the compound Examples C = C ; – C ≡ C –; – C ≡ N; – N = N –; C = O; etc., Chromophores undergo $\pi \rightarrow \pi^*$ transitions in the short wavelength regions of UV-radiations.

Auxochromes: Auxochrome is a saturated group (containing lone pair of electron), when auxochrome attach to the Chromophore absorption shift towards longer wavelength. For example, -OH, -SH, -NH etc.



Absorption, intensity shift & UV spectrum:

1) **Bathochromic shift**: It is also known as **Red shift,** in this case absorption shift towards **longer wavelength (λ_{max}).**
2) **Hypsochromic shift**: It is also known as **Blue shift,** in this case absorption shift towards **shorter wavelength (λ_{max}).**
3) **Hyperchromic shift:** Intensity of absorption maximum (ε_{max}) increases.
4) **Hypochromic shift:** Intensity of absorption maximum (ε_{max}) decrease



IR Spectroscopy Defination, Principle and IR Spectroscopy Instrumentation

IR Spectroscopy :-

Absorption of Light in IR Region Causes Transition Between Vibrational & Rotational Energy levels So it is also Called Vibrational and Rotation spectroscopy

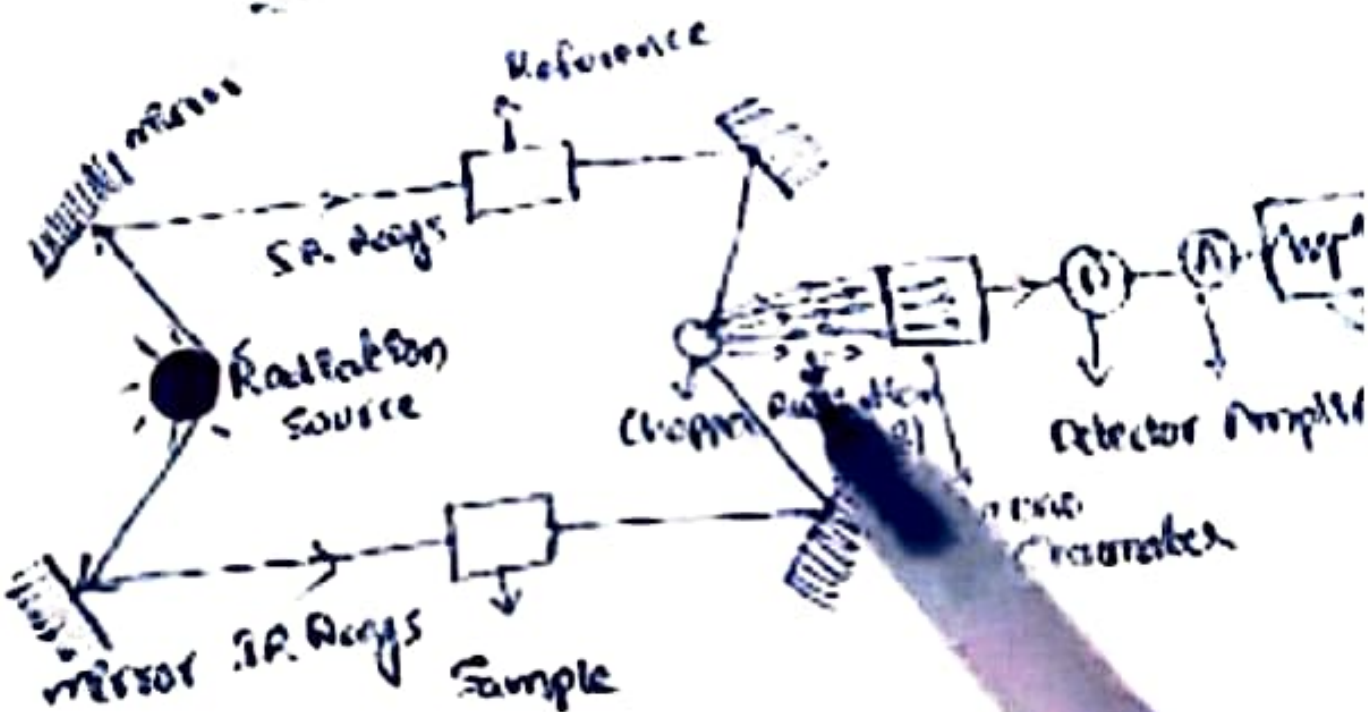
principle:- IR Active Compounds we get IR Spectrum gets Excited & Shows Vibrational & rotational spectra which characterstic for functional groups & finger print region

Near IR	MID IR	Far IR
14000 – 4000 cm^{-1}	4000 – 400 cm^{-1}	400 – 10 cm^{-1}

Finger print region = 625 – 1500 cm^{-1}

Functional group = 1500 cm^{-1} – 4000 cm^{-1}

Instrumentation of I.R. spectroscopy



Thermo Couple:-

Temparature difference, potential difference

Thermister:- fused metal oxide mixture

Temp ↑s & ↓ resistence

Golay detector:- Xe - is filled in cylinder

Photo Conductivity Detector:- pbs is present

Amplifier:- Amplifier Amplifies Signal from detector

Applications of IR:- Chamber and recorder record the signal which is displayed read out

1. identification of Substance
2. Determination of Molecular structure
3. ~~Sty~~ Studying progress of Reactions
4. Detection of impurities

   Ex: Ketone in alcohol

① Radiation Source

a) nearnest glover :-

oxides of zirconium, yittrium, Erbium

(b) Glober : Rod of sintered silicon Carbide

② Sampling Technique:

1. solid Sampling
2. Liquid "
3. Gas "
4. solution

③ Monochromator:-

prisims - Alkali metal halides like NaCl, KBr, LiF, CeBr

Grating: A Set of Equally spaced paralleled wires used to produce spectra by using diffraction

④ Detectors:

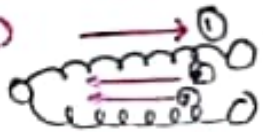
Bolometer Thin metal Conductor

Temp ↑s & ↑s Resistence → Current flows Changes

Functional Group Region.

Two types of Fundamental Vibrations:-

* Stretching
  - Symmetrical
    - cale to either move away or toward
  - Asymmetrical

① 

Same side
Two atoms of mole

The Central atom
opposite side

②

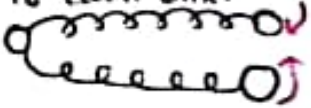
Symmetrical Asymmetrical

* One atom approch towards Central atom while depart from it

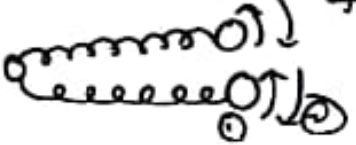
* Bending → four types

I) In plane Bending

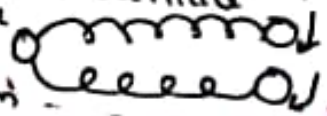
① Scissoring
In plane bending occurs wherein They move back & forward They approch to each other



② Rocking
In plane Bending of atoms occurs where in They move back ward & forward with respect to central atom



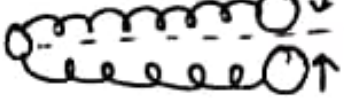
③ wagging
Two atoms oscillate up and Below the plane respect to central atom



plane

Bond angle Same

④ Twisting = one of atom move up the plane while other downy



plane the pl
w.r. to Centra

5 Isomerisim in organic chemistry

6 Identification of Functional groups
presence of water in sample

7 Biomedical studies of human hair
to identify disease

8 identify air pollutants & water
identify odour & taste components
of food

Spectroscopy is also known as ***Vibrational spectroscopy***

Selection Rule for IR :

Requirement for a molecule to impart IR spectrum

The essential requirement for a molecule to show vibrational spectrum is that the dipole moment of the molecule must change during vibration. This type of spectra is given by heteronuclear diatomic molecules, and poly atomic molecules with and without dipole moment. In polyatomic molecule, having no dipole moment, the exciting radiation can induce an oscillating dipole moment, hence it becomes IR active. Homonuclear diatomic molecules such as O_2, N_2, H_2, etc., are IR inactive because they have no dipole moment. The vibration of a diatomic molecule may be considered as the vibration of a simple harmonic oscillator. The vibrational energy of a diatomic molecule is given as $E_{vib} = (v + 1/2)\, h\, \nu$, where, v = the vibrational quantum number 0, 1, 2, 3 etc., and ω = fundamental vibrational frequency. $\nu = 1/2\pi\, (k/\mu)^{1/2}$, k = force constant and μ = reduced mass.

When a molecule is in the ground state ($V = 0$), $E_{vib} = 1/2\, h\, \nu$. This is the energy of the lowest vibrational level called the zero point energy. The selection rule is $\Delta E = \pm 1$, $\Delta E_{vib} = h\, \nu$, i.e., For a vibrational transition

4. *Finger print region*

The vibrational spectral (*IR* spectra) region at 1500 — 600 cm⁻¹ ~~1400 — 700~~ cm^{-1} gives very rich and intense absorption bands. This region is termed as finger print region. The region 4000 — 1500 ~~1400~~ cm^{-1} is known as **Group frequency region**.

Uses of finger print region

Finger print region can be used to detect the presence of functional group and also to identify and characterise the molecule just as a finger print can be used to identify a person.

Fermi resonance

1. When a fundamental vibration couples with an overtone or combination band, the coupled vibration is called Fermi resonance

Fermi resonance is often observed in carbonyl compounds

2. Fermi resonance results in the splitting of two vibrational bands that have nearly the same energy and symmetry in both IR and Raman spectroscopies

3. The two bands are usally a fundamental vibration and either an overtone or combination band

Overtones :-

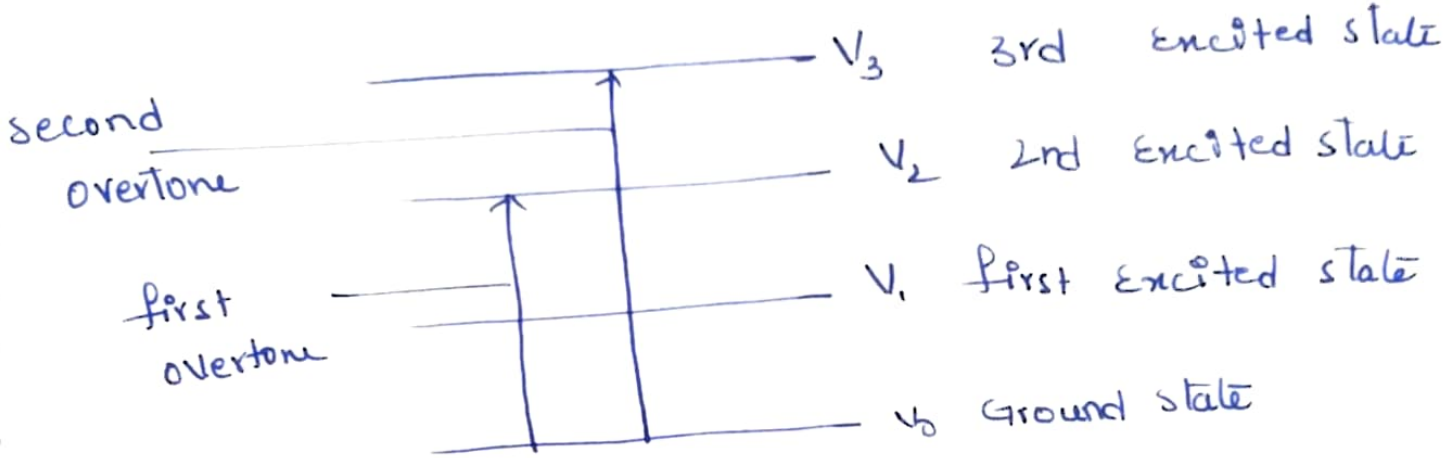
The transition occurs from ground state to second excited state through first excited state (fundamental vibration) is called overtone

overtone occurs when a vibrational mode is excited from $v = 0$ to $v = 2$ which is called first overtone (or) $v = 0$ to $v = 3$ the second overtone. The fundamental transitions, $v = \pm 1$ are the most occuring

*** The probability of overtone rapid decreases as the no. of quanta "$\Delta v = \pm n$" increases

Excitation from ground state to higher Energy Excited state i.e 2nd or 3rd observed at twice of frequency of a strong band

Results from Excited from ground state to the 2nd or 3rd Excited state and Correspond to integral Multiple of the frequency of the ~~fundal mental~~ fundamental band



$V_0 - V_2$ = first overtone

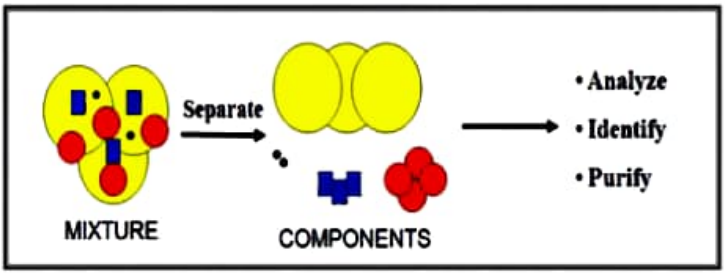
$V_0 - V_3$ = second overtone

$V_0 - V_1$ = fundamental Vibration

✦ CHROMATOGRAPHY

Chromatography is the most important analytical techniques used widelyand frequently for separation, purification and identification of chemical components present in the complex mixture which connot separate using physical and chemical methods.The word chromatography means "color writing" which is a way that a chemist can test liquid mixtures., M.S. Tswett , a Russian botanist invented chromatography in 1903, while studying the coloring materials in plant life.

Chromatography : **Chromatography is a laboratory technique** for separating the components of a mixture on the basis of differences in their affinity for **a stationary and a mobile phase** in order to analyze, identify, purify, and/or quantify the mixture or components.



Chromatography involved two mutually immiscible phases they are:

1) **Stationary phase** : The stationary phase in chromatography is the fixed phase that is either a solid or liquid particle attached to a glass or a metal surface on which the components of the mixture to be separated is absorbed selectively.

   Ex : **Silica gel, alumina and cellulose powder**

2) **Mobile phase** : The mobile phase in chromatography is the phase that is either liquid or gas that is passed through a chromatographic system where the components of the mixture are separated at different raters by adsorbing them to the stationary phase.

   Ex: **Gas, liquid or supercritical fluid**

Classofication of Chromtography :

There are different types of chromatography based on the type of stationary phase and mpbile phase used . They are

1) Gas- Solid Chromatography
2) Gas- liquid Chromatography
3) Solid - Liquid chromatography : Column , TLC and HPLC
4) Liquid- Liquid Chromatography : Paper, Coloumn partition chromatography.

Types of Chromatography Techniques :

1) Paper Chromatography
2) Thin layer chromatography (TLC)
3) Column Chromatography
4) Gas Chromatography (GC)
5) High Performance Liquid Chromatography (HPLC)
6) Ion Exchange Chromatography
7) Affinity Chromatography
8) Gel chromatography

High Performance Liquid Chromatography

High perfomance Liquid Chromatography

High performance liquid chromatography (HPLC) is basically a highly improved form of column liquid chromatography.

Instead of a solvent being allowed to drip through a column under gravity, it is forced through under high pressures of up to 400 atmospheres. That makes it much faster.

All chromatographic separations, including HPLC operate under the same basic principle; separation of a sample into its constituent parts because of the difference in the relative affinities of different molecules for the mobile phase and the stationary phase used in the separation.

Types of HPLC

There are following variants of HPLC, depending upon the phase system (stationary) in the process :

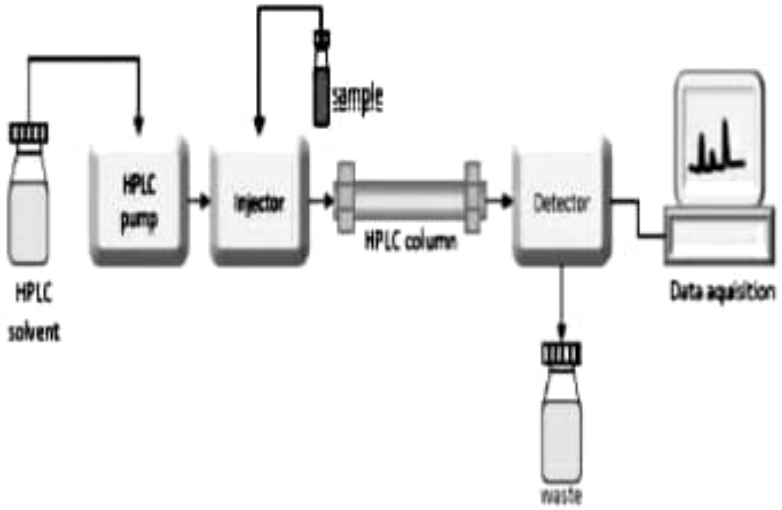
1. Normal Phase HPLC

This method separates analytes on the basis of polarity. NP-HPLC uses polar stationary phase and non-polar mobile phase. Therefore, the stationary phase is usually silica and typical mobile phases are hexane, methylene chloride, chloroform, diethyl ether, and mixtures of these.

Polar samples are thus retained on the polar surface of the column packing longer than less polar materials.

2. Reverse Phase HPLC

The stationary phase is nonpolar (hydrophobic) in nature, while the mobile phase is a polar liquid, such as mixtures of water and methanol or acetonitrile. It works on the principle of hydrophobic interactions hence the more nonpolar the material is, the longer it will be retained.



1. Solvent Resorvoir

Mobile phase contents are contained in a glass resorvoir. The mobile phase, or solvent, in HPLC is usually a mixture of polar and non-polar liquid components whose respective concentrations are varied depending on the composition of the sample.

2. Pump

A pump aspirates the mobile phase from the solvent resorvoir and forces it through the system's column and detecter. Depending on a number of factors including column dimensions, particle size of the stationary phase, the flow rate and composition of the mobile phase, operating pressures of up to 42000 kPa (about 6000 psi) can be generated.

3. Sample Injector

The injector can be a single injection or an automated injection system. An injector for an HPLC system should provide injection of the liquid sample within the range of 0.1-100 mL of volume with high reproducibility and under high pressure (up to 4000 psi).

4. Columns

Columns are usually made of polished stainless steel, are between 50 and 300 mm long and have an internal diameter of between 2 and 5 mm. They are commonly filled with a stationary phase with a particle size of 3–10 μm.

Columns with internal diameters of less than 2 mm are often referred to as microbore columns. Ideally the temperature of the mobile phase and the column should be kept constant during an analysis.

5. Detector

The HPLC detector, located at the end of the column detect the analytes as they elute from the chromatographic column. Commonly used detectors are UV-spectroscopy, fluorescence, mass-spectrometric and electrochemical detectors.

6. Data Collection Devices

Signals from the detector may be collected on chart recorders or electronic integrators that vary in complexity and in their ability to process, store and reprocess chromatographic data. The computer integrates the response of the detector to each component and places it into a chromatograph that is easy to read and interpret.

Applications of HPLC

The information that can be obtained by HPLC includes resolution, identification and quantification of a compound. It also aids in chemical separation and purification. The other applications of HPLC include :

Pharmaceutical Applications

1. To control drug stability.

2. Tablet dissolution study of pharmaceutical dosages form.

3. Pharmaceutical quality control.

Environmental Applications

1. Detection of phenolic compounds in drinking water.

2. Bio-monitoring of pollutants.

Applications in Forensics

1. Quantification of drugs in biological samples.

2. Identification of steroids in blood, urine etc.

3. Pharmaceutical quality control.

Environmental Applications

1. Detection of phenolic compounds in drinking water.

2. Bio-monitoring of pollutants.

Applications in Forensics

1. Quantification of drugs in biological samples.

2. Identification of steroids in blood, urine etc.

3. Forensic analysis of textile dyes.

4. Determination of cocaine and other drugs of abuse in blood, urine etc.

Food and Flavour

1. Measurement of Quality of soft drinks and water.

2. Sugar analysis in fruit juices.

3. Analysis of polycyclic compounds in vegetables.

4. Preservative analysis.

Applications in Clinical Tests

1. Urine analysis, antibiotics analysis in blood.
2. Analysis of bilirubin, biliverdin in hepatic disorders.
3. Detection of endogenous Neuropeptides in extracellular fluid of brain etc.